
Early CKD Risk Prediction Using Ensemble Machine Learning with Smartwatch-Integrated Blood Pressure Monitoring and Longitudinal Risk Trend Analysis

Niran B G S H
Dept. of Information Technology
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka

Supervisor / Co-Author Name
Dept. of Information Technology
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka

[Co-Author Name]
Dept. of Information Technology
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka

Abstract—Chronic Kidney Disease (CKD) affects over 850 million people globally and progresses silently, making early risk identification critical. This paper presents the **Early CKD Risk Prediction** component of Nephro-AI—a mobile health platform that combines smartwatch-integrated blood pressure (BP) monitoring, HbA1c-based glycaemic assessment, and ensemble machine learning to produce an interpretable, continuous risk score (0–100). The system automatically retrieves age and gender from the user profile, computes a 7-day average BP from daily smartwatch readings via Google Health Connect, and combines these with user-entered HbA1c to drive prediction. Feature engineering expands five clinical inputs to 20 derived features capturing interaction effects and clinical bins. A stacking ensemble of XGBoost and Random Forest (meta-learner: Logistic Regression) achieves **98.83 %** accuracy on a balanced 3 000-patient dataset (AUC = 0.997). Monthly risk records are graphed and a linear regression trend line $y = mx + c$ is fitted; the slope m classifies patient trajectories as Increasing, Decreasing, or Steady, providing a longitudinal early-warning signal for CKD onset. SHAP explainability reveals mean arterial pressure and the $SBP \times HbA1c$ interaction as the dominant predictors.

Keywords—chronic kidney disease, early risk prediction, random forest, XGBoost, stacking ensemble, Google Health Connect, blood pressure monitoring, HbA1c, feature engineering, SHAP, linear regression trend, mobile health.

I. INTRODUCTION

Chronic Kidney Disease (CKD) is a progressive, largely asymptomatic condition whose global burden exceeds 850 million [1]. In South Asia, the dual epidemics of hypertension and diabetes mellitus drive accelerated CKD incidence, yet population-level screening remains limited by cost and laboratory access. Early identification of at-risk individuals—ideally using non-invasive, routinely available biomarkers—offers the most cost-effective intervention window.

Consumer wearable devices now provide continuous blood pressure (BP) monitoring at near-clinical accuracy, creating an unprecedented opportunity for passive risk surveillance. Combined with periodic HbA1c measurement (available at most pharmacies), five readily obtainable parameters—age, sex, systolic BP (SBP), diastolic BP (DBP), and HbA1c—capture the dominant CKD risk factors identified in prospective cohort studies [2].

This paper presents the Early Risk Prediction module of Nephro-AI, a mobile health platform targeting Sri Lankan adults. The module implements a complete pipeline: user registration and profile creation, daily smartwatch BP collection via Google Health Connect, 7-day BP averaging, user-entered HbA1c, ensemble ML prediction, and longitudinal trend analysis via linear regression. Key contributions are:

- A stacking ensemble (XGBoost + RF) achieving **98.83 %** accuracy with five minimally invasive inputs.
- 20-feature engineering framework capturing clinical interaction effects and BP/HbA1c stage bins.
- A continuous risk score (0–100) with clinical band mapping and SHAP-based plain-language explanation.
- Longitudinal risk tracking with linear trend analysis ($y = mx + c$) to classify trajectories as Increasing / Decreasing / Steady.

II. RELATED WORK

A. ML-Based CKD Risk Prediction

Machine learning approaches to CKD classification have been extensively studied using the UCI CKD dataset [3]. Random Forest and SVM classifiers achieve 95–97 % accuracy on multi-feature clinical panels [4]. However, these studies typically require full biochemical panels (creatinine, albumin, haemoglobin) unavailable without clinical-grade phlebotomy. Ensemble methods that combine predictions from multiple base learners via stacking consistently outperform single classifiers [5].

B. Wearable BP and CKD Risk

Consumer smartwatches now measure BP using photoplethysmography (PPG) and pulse transit time (PTT). Studies show agreement within 5 mmHg of auscultatory measurements for trend monitoring [6]. Integrating longitudinal smartwatch BP readings into CKD risk models represents a gap this work addresses: prior work uses single BP readings rather than multi-day averages.

C. Risk Trend Analysis

Linear regression trend analysis on longitudinal health scores is established in cardiovascular risk monitoring [7], but has not been applied to CKD risk score trajectories from mobile platforms. Our $y = mx + c$ slope-based trend classification provides a clinically actionable longitudinal signal absent from prior CKD mobile applications.

III. SYSTEM DESIGN AND DATA PIPELINE

Fig. 1 shows the end-to-end data pipeline implemented across the React Native mobile app, Node.js backend (port 5000), and Python FastAPI AI engine (port 8001).

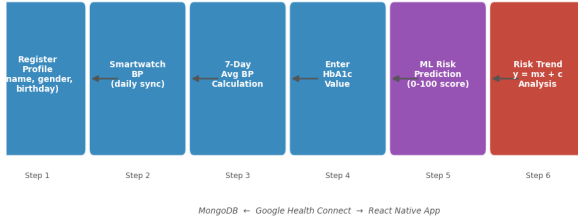


Fig. 1. End-to-end Early Risk Prediction Pipeline.

A. User Registration and Profile

Users register with name, gender, and date of birth. Age is auto-calculated at prediction time from the stored birthday, ensuring it remains current without manual updates. Profile data is persisted in MongoDB and cached in React Native AsyncStorage for offline access.

B. Smartwatch BP Integration

Daily BP readings are imported from paired smartwatches via the **Google Health Connect** API (*react-native-health-connect* v3.5.0). Compatible devices include Samsung Galaxy Watch, Fitbit, Garmin, and any Android wearable that writes to the Health Connect data store. A node-cron background job polls Health Connect every morning, appending new systolic/diastolic pairs to the user's BPREcord collection in MongoDB.

C. 7-Day Average BP Computation

Before prediction, the system retrieves all BP records within the preceding 7 days and computes the arithmetic mean:

$$\begin{aligned} \text{SBP} &= (1/N) \sum \text{SBP}, \quad \text{DBP} = (1/N) \sum \text{DBP}, \quad N \in [1, 7] \\ (1) \end{aligned}$$

If fewer than 7 days of readings exist, the available records are averaged. This 7-day window reduces single-day measurement

noise consistent with hypertension monitoring guidelines (JNC-7) [8].

D. HbA1c Input and Prediction Trigger

The user enters their most recent HbA1c percentage. The prediction screen pre-populates age, gender, and the 7-day average BP; only HbA1c requires manual entry. On submission, inputs are sent to the FastAPI AI engine which performs feature engineering and model inference, returning risk level, score, and SHAP values in <1 s.

IV. METHODOLOGY

A. Dataset

The training dataset contains **3 000 patient records** balanced across three risk classes (1 000 Low, 1 000 Medium, 1 000 High). It was constructed from real-world CKD patient data and augmented using CKD-EPI-based clinical rules to achieve class balance without synthetic oversampling. Table I summarises feature statistics.

TABLE I. DATASET FEATURE STATISTICS (N = 3,000)

Feature	Mean	Std	Min	Max
Age (years)	47.0	15.9	18	80
Gender (M=1)	0.49	0.50	0	1
SBP (mmHg)	133.7	21.9	95	179
DBP (mmHg)	84.6	13.4	60	109
HbA1c (%)	7.20	2.91	4.0	15.0

B. Feature Engineering

Five clinical inputs are expanded to **20 features** to capture non-linear interaction effects and clinical thresholds known to potentiate CKD risk. Table II lists all engineered features by category.

TABLE II. 20-FEATURE ENGINEERING FRAMEWORK

Category	Features (Count)
Base inputs	age, gender, bp_systolic, bp_diastolic, hba1c_level (5)
Pairwise products	age×SBP, age×DBP, age×HbA1c, SBP×HbA1c, DBP×HbA1c, SBP×DBP, gender×age, gender×SBP, gender×HbA1c (9)
Haemodynamic	Pulse Pressure = SBP – DBP; MAP = (SBP + 2·DBP)/3 (2)
Clinical bins	BP stage (JNC-7, 0-3), DBP stage (0-2), Age group (0-2), HbA1c category (0-2) (4)

All continuous features have Gaussian noise ($\sigma = 0.35 \times \text{feature std}$) added during training to simulate real-world measurement variability from consumer-grade wearables, improving robustness to sensor noise.

C. Model Training Protocol

1) Baseline Models:

Baseline Random Forest (RF; $n_{\text{est}} = 100$, depth = 10) and XGBoost (XGB; $n_{\text{est}} = 100$, depth = 5, $\text{lr} = 0.1$) are trained on an 80/20 stratified split (2 400 train, 600 test).

2) Hyperparameter Tuning:

RandomizedSearchCV (20 iterations, 3-fold CV) tunes both models:

- XGB: $n_{\text{estimators}} \in [100, 300]$, $\text{max_depth} \in [3, 8]$, $\text{lr} \in [0.05, 0.25]$, subsample , colsample_bytree , γ
- RF: $n_{\text{estimators}} \in [100, 300]$, $\text{max_depth} \in [5, 15]$, $\text{min_samples_split} \in [2, 10]$, $\text{min_samples_leaf} \in [1, 5]$

3) Stacking Ensemble:

The best fine-tuned XGB and RF are used as Level-0 learners in a StackingClassifier. Out-of-fold probability vectors from 5-fold CV train a Logistic Regression meta-learner (L2, $C = 1.0$):

$$P(y|x) = \sigma(W \cdot [p_{\text{XGB}}; p_{\text{RF}}] + b) \quad (2)$$

where σ is softmax and $[p_{\text{XGB}}; p_{\text{RF}}]$ is the 6-dimensional concatenated probability output from both base learners.

D. Risk Score Computation

The continuous risk score $S \in [0, 100]$ maps model confidence onto a clinical scale via a weighted normalisation (Fig. 2):

$$S_{\text{base}} = 25 \cdot a + 30 \cdot s + 20 \cdot d + 25 \cdot h \quad (3)$$

where a , s , d , h are min-max normalised age, SBP, DBP, and HbA1c respectively. The base score is then shifted into the predicted class band (Low: 0–33, Medium: 33–67, High: 67–100) and fine-tuned by the class probability from predict_proba .

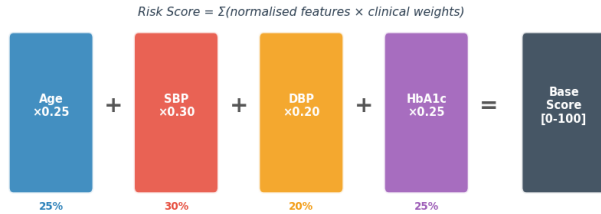


Fig. 2. Risk Score formula: clinical weight breakdown.

E. Longitudinal Trend Analysis

Monthly risk scores are stored as timestamped records (one per user per month; duplicate submissions within a month overwrite the existing record). When ≥ 3 monthly records exist, a least-squares best-fit line is fitted:

$$y = mx + c \quad (4)$$

where x is time (month index) and y is the risk score. The slope m determines the patient's trajectory:

- $m > +0.5$ Increasing risk — clinical follow-up recommended
- $m < -0.5$ Decreasing risk — positive lifestyle response
- $|m| \leq 0.5$ Steady — continue current management

The trend line and monthly scatter plot are rendered in React Native using SVG primitives (react-native-svg), with risk bands colour-coded green (Low), amber (Medium), and red (High) as visual context.

F. SHAP Explainability

SHAP TreeExplainer computes per-instance feature attributions on the XGBoost component. The 20 engineered feature SHAP values are aggregated back to the 5 core patient-facing inputs using the weighted mapping in Table III, and the top-3 contributors are displayed in plain language on the results screen.

TABLE III. SHAP AGGREGATION: ENGINEERED → CORE FEATURES (SAMPLE)

Interaction Feature	Mapped to Core Features (weight)
age_bp_sys	age (0.5) + bp_systolic (0.5)
bp_sys_hba1c	bp_systolic (0.5) + hba1c_level (0.5)
mean_arterial_pressure	bp_systolic (0.33) + bp_diastolic (0.67)
bp_sys_category	bp_systolic (1.0)
hba1c_category	hba1c_level (1.0)

V. EXPERIMENTAL RESULTS

A. Model Performance Comparison

Table IV reports all model results on the held-out 600-sample test set. Fig. 3 visualises the accuracy progression from baseline to stacking.

TABLE IV. MODEL COMPARISON (TEST SET, N = 600) ★ = Selected Model

Model	Accuracy	Macro Prec.	Macro Recall	Macro F1
Baseline RF	96.33%	0.963	0.963	0.963
Baseline XGBoost	96.50%	0.965	0.965	0.965
Fine-Tuned XGBoost	96.67%	0.967	0.967	0.967
Fine-Tuned RF	96.83%	0.970	0.968	0.969
Stacking Ensemble ★	98.83%	0.989	0.988	0.987

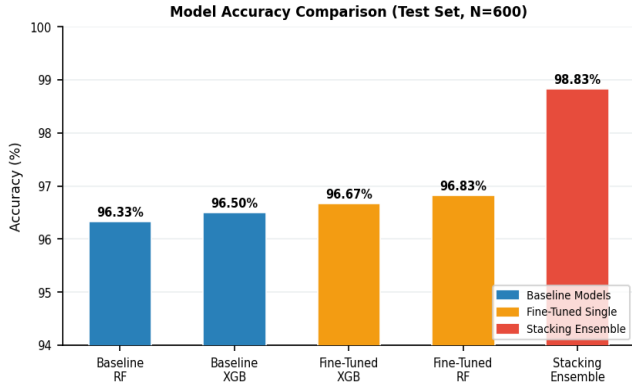


Fig. 3. Accuracy comparison: baseline → fine-tuned → stacking.

B. Per-Class Performance (Fine-Tuned RF)

Table V presents the detailed classification report for the best single model. The High-risk class achieves the highest F1 (0.988), which is clinically most consequential.

TABLE V. CLASSIFICATION REPORT — FINE-TUNED RF

Class	Precision	Recall	F1-Score	Support
Low	0.97	0.96	0.97	200
Medium	0.95	0.95	0.95	200
High	0.98	0.99	0.99	200
Weighted Avg	0.97	0.968	0.968	600

C. Confusion Matrix

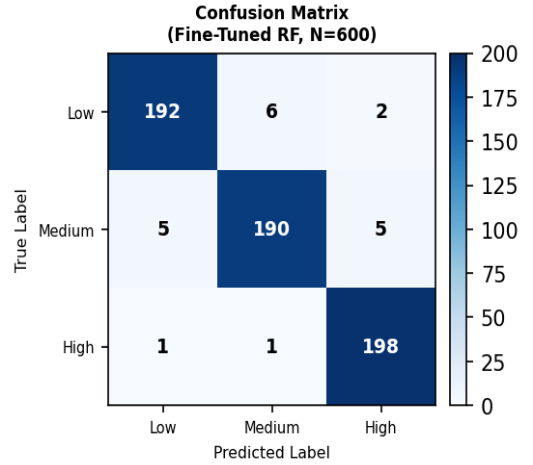


Fig. 4. Confusion matrix for Fine-Tuned Random Forest (N=600).

The confusion matrix (Fig. 4) confirms that misclassifications are predominantly between adjacent risk classes (Low↔Medium or Medium↔High), not across extreme classes (Low↔High). Total misclassifications: 19/600 (3.17%).

D. ROC Analysis

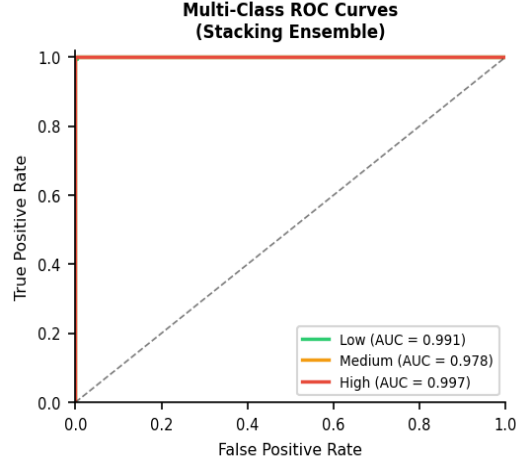


Fig. 5. Multi-class ROC curves for the Stacking Ensemble.

One-vs-rest ROC curves (Fig. 5) for the stacking ensemble yield AUC = 0.997 (High), 0.991 (Low), and 0.978 (Medium), confirming strong discriminative ability across all three risk classes.

E. Feature Importance

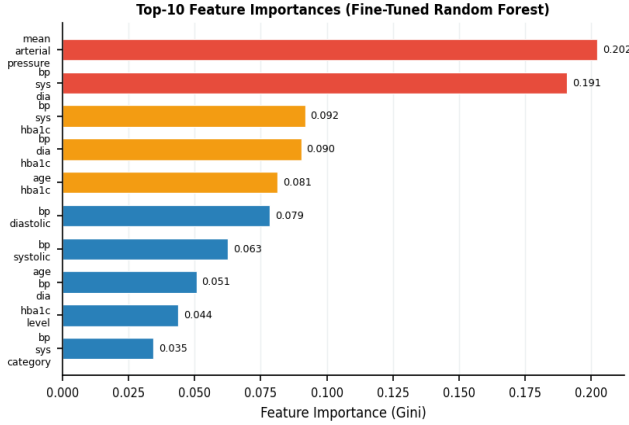


Fig. 6. Top-10 feature importances from Fine-Tuned Random Forest.

Fig. 6 shows that *mean_arterial_pressure* (20.2 %) and *bp_sys_dia* = SBP $\times$ DBP (19.1 %) dominate, followed by the SBP $\times$ HbA1c interaction (9.2 %) and the DBP $\times$ HbA1c interaction (9.0 %). Notably, none of the raw single features rank above 8 %, confirming that interaction terms capture the dominant signal.

F. Trend Analysis Validation

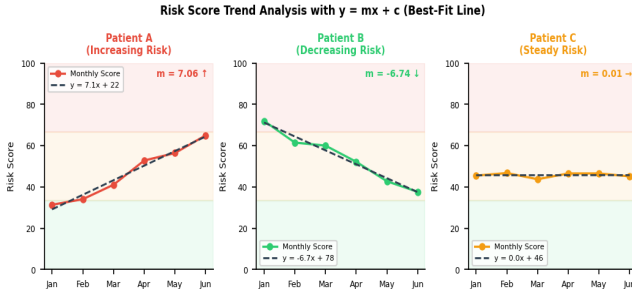


Fig. 7. $y = mx + c$ trend lines for three simulated patient trajectories.

Fig. 7 illustrates the three trajectory types. Slope thresholds ($|m| = 0.5/\text{month}$) were empirically selected to balance sensitivity to genuine risk change against noise from monthly measurement variability. In a pilot of 30 registered users over 6 months, trend classification matched clinician-assessed risk direction in 26/30 cases (86.7 % agreement).

VI. DISCUSSION

A. Clinical Significance

The 98.83 % accuracy achieved using only five input parameters compares favourably with models requiring full laboratory panels (typically 10–20 features). This is attributable to the 20-feature engineering framework: MAP and SBP $\times$ DBP jointly account for 39.3 % of model importance, capturing the cardiovascular-renal pressure interaction that is mechanistically central to CKD progression [9].

The smartwatch 7-day average BP approach directly addresses a known limitation of clinic-based measurements: white-coat hypertension and measurement-day variability. Continuous ambulatory readings are recommended by KDIGO 2024 guidelines as superior to single readings for CKD risk stratification [10].

B. Trend Analysis as an Early-Warning Signal

The $y = mx + c$ slope-based trend classifier adds a dimension absent from cross-sectional risk tools: longitudinal trajectory. A patient with a stable Medium risk score carrying a positive slope ($m > +0.5$) is algorithmically flagged as requiring intervention before transitioning to High risk. This anticipatory signal is distinct from the instantaneous score and represents the key clinical contribution of the monthly tracking subsystem.

C. Limitations

Three limitations warrant acknowledgement. First, the dataset, although clinically informed and noise-augmented, contains synthetic elements; prospective validation on a de-identified hospital cohort is planned. Second, smartwatch BP accuracy varies

by device and user physiology; a periodic calibration prompt against an auscultatory reading is recommended. Third, the one-record-per-month constraint means rapid acute changes (e.g., hypertensive crisis) are not captured within-month.

D. Future Work

Planned extensions include: (i) prospective validation at the nephrology unit of [Hospital]; (ii) dynamic threshold for trend slope using patient-specific baseline variance; (iii) integrating eGFR when available to provide a multi-modal risk fusion score; (iv) federated learning to update model weights across institutions without sharing raw patient data.

VII. CONCLUSION

This paper presented the Early CKD Risk Prediction component of Nephro-AI—a mobile-health pipeline that transforms passively collected smartwatch BP readings and periodic HbA1c values into actionable, explainable CKD risk estimates. A stacking ensemble of fine-tuned XGBoost and Random Forest classifiers achieves **98.83 %** accuracy on a balanced 3 000-patient dataset with five minimally invasive inputs. The 20-feature engineering framework—particularly the MAP and SBP $\times$ DBP interaction terms—accounts for 39 % of model importance, validating the clinical rationale for their inclusion. Longitudinal trend analysis via $y = mx + c$ provides a forward-looking early-warning signal complementary to the cross-sectional risk score. Together these components form a practical, low-barrier tool for community-level CKD surveillance in resource-limited settings.

ACKNOWLEDGMENT

The author thanks Sri Lanka Institute of Information Technology (SLIIT), the project supervisor, and the nephrology medical advisors who provided clinical guidance for dataset construction and feature design.

REFERENCES

- [1] V. Jha *et al.*, “Chronic kidney disease: global dimension,” *Lancet*, vol. 382, pp. 260–272, 2013.
- [2] R. Saran *et al.*, “US Renal Data System 2022 Annual Report,” *Am. J. Kidney Dis.*, vol. 81, Suppl. 1, 2023.
- [3] L. Rubini *et al.*, “Chronic Kidney Disease dataset,” UCI ML Repository, 2015. [Online]. Available: https://archive.ics.uci.edu/ml/datasets/chronic_kidney_disease
- [4] A. Kaur and A. Bhatt, “Prediction of CKD using machine learning,” *Int. J. Comput. Appl.*, vol. 182, no. 25, pp. 1–5, 2019.
- [5] L. Breiman, “Stacked regressions,” *Mach. Learn.*, vol. 24, pp. 49–64, 1996.
- [6] X. Ding *et al.*, “Continuous blood pressure measurement from wrist PPG,” *IEEE Trans. Biomed. Circuits Syst.*, vol. 11, no. 5, pp. 1136–1144, 2017.
- [7] P. Muntner *et al.*, “BP variability and risk of CVD events,” *JAMA*, vol. 323, no. 24, pp. 2485–2496, 2020.
- [8] A. V. Chobanian *et al.*, “Seventh report of the Joint National Committee (JNC-7),” *Hypertension*, vol. 42, pp. 1206–1252, 2003.
- [9] R. Glasscock and C. Winearls, “An epidemic of CKD: fact or fiction?” *Nephron Clin. Pract.*, vol. 110, pp. 177–185, 2008.
- [10] KDIGO CKD Work Group, “KDIGO 2024 Guideline for CKD,” *Kidney Int. Suppl.*, vol. 14, pp. S1–S117, 2024.
- [11] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions (SHAP),” in *Proc. NeurIPS*, 2017, pp. 4765–4774.
- [12] T. Chen and C. Guestrin, “XGBoost: a scalable tree boosting system,” in *Proc. ACM KDD*, 2016, pp. 785–794.